

Session 1 : Introduction to Artificial Intelligence

Welcome to your AI journey! Today we explore what Artificial Intelligence really means and discover how it's already a part of your daily life, from the shows you watch to the apps you use.

Get ready to explore the world of AI 🚀





Today's Objective

By the end of this session, you will be able to:

1

Define AI

Explain what Artificial Intelligence means in your own words

2

Spot AI Around You

Identify AI tools and apps you already use every single day

3

Understand the Framework

Explain how AI works using the **Sense → Think → Act** model

4

Bust the Myths

Tell the difference between facts and common myths about AI

5

AI = Tool, Not Replacement

Understand that AI is a helpful tool – not a replacement for human thinking



What Do You Already Know About AI?

Before we dive in, let's warm up our brains! Take a moment to think about these questions and share your thoughts with the group:

🤔 First Thoughts

What comes to mind when you hear "Artificial Intelligence"? Robots? Smart phones? Something else?

📱 Smart Devices

Have you used any apps or devices that seemed "smart" — like they knew what you wanted?

? Burning Questions

What questions do you have about AI? There are no silly questions here — every question is a great one!

Take a moment to share your thoughts! You might be surprised by how much you already know. 💡

What is Artificial Intelligence?

Artificial Intelligence (AI) is a branch of computer science focused on creating machines and programs that can perform tasks that would normally require human intelligence. **Think of it this way:** scientists taught computers to do some of the clever things our brains do!

These tasks include:



Learning

Getting better from experience, just like you do with practice



Recognizing Patterns

Spotting similarities and trends in huge amounts of information



Understanding Language

Reading and responding to words we write or speak



Making Decisions

Choosing the best option based on what it has learned



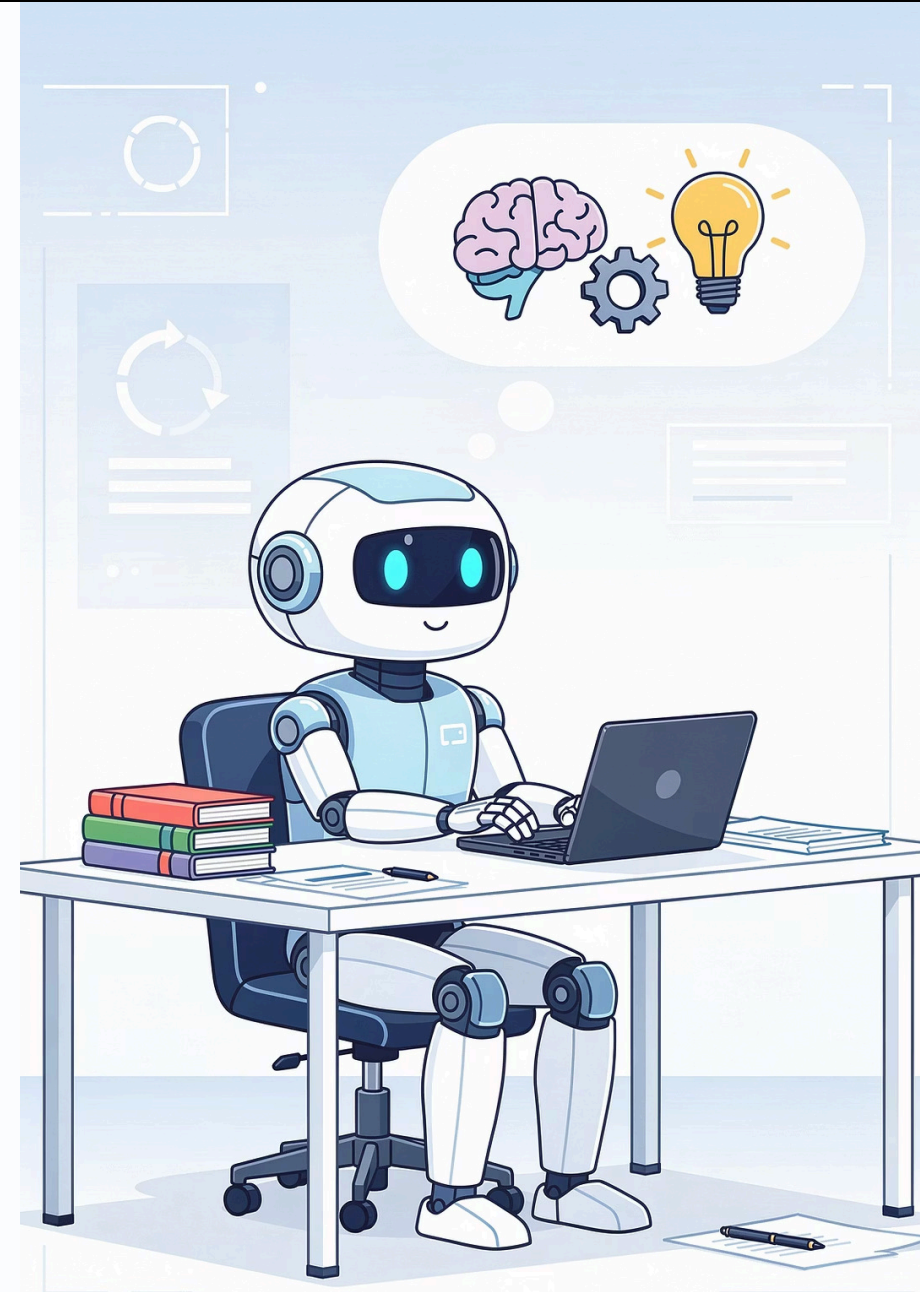
Solving Problems

Working through challenges step by step



Creating Content

Generating new text, images, and even music



AI = Patterns, Not Magic

This is one of the **most important things** to understand about AI:

AI Learns From Examples

People feed a computer thousands, sometimes **millions** of examples. For instance, to recognize cats, the AI might look at 10 million cat photos.

AI Finds Patterns

The computer notices what all those cat photos have in common: pointy ears, whiskers, certain shapes. It builds an internal "checklist" of cat features.

AI Makes Predictions

When you show it a new photo, it checks its pattern list and says, "This looks 94% like a cat!" It's making a **guess** based on what it learned.

❏ ⚠️ **Common Mistake:** Some people think AI "understands" cats the way you do. It doesn't! It has never pet a cat, heard a cat purr, or felt a cat's soft fur. It only knows numbers and patterns in pixels.

AI vs. Traditional Programs

What makes AI special? Let's compare it to the regular programs you already know!

Traditional Programs

- Follow **fixed rules** written by programmers
- Cannot learn or adapt on their own
- Always do the exact same thing

 **Example:** A calculator always adds numbers the same way. $2 + 2 = 4$, every single time. No surprises!

AI-Powered Systems

- **Learn** from data and examples
- **Improve** over time with more practice
- Can handle new situations they haven't seen before

 **Example:** A spam filter learns to identify unwanted emails by studying thousands of examples — and gets smarter every day!

 **The key difference:** AI *learns and adapts*; traditional programs just *follow instructions*.

AI in Everyday Life

One of the most exciting things about AI is that it's already part of our daily routines – often without us even realizing it.



Netflix / Prime Video / Disney+

These streaming services use AI to analyze what you've watched and recommend other shows or movies you might enjoy. The more you watch, the better the recommendations get.



YouTube

Similar to Netflix, YouTube's recommendation engine suggests videos based on your viewing history, likes, and searches. It's why you might fall down a rabbit hole of related content.



Siri / Google Assistant / Alexa

Voice assistants use AI to understand your spoken words, interpret what you mean, and respond with useful information or actions. They learn from your speech patterns over time.

AI in Everyday Life



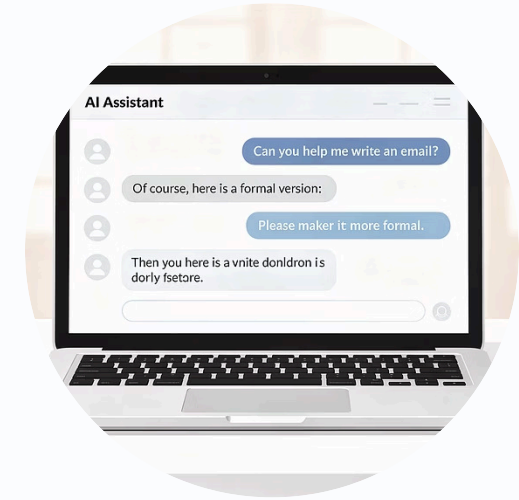
Google Maps

When you ask for directions, Google Maps uses AI to predict traffic conditions, estimate travel time, and suggest the fastest route. It learns from real-time data from millions of users.



Instagram / Snapchat

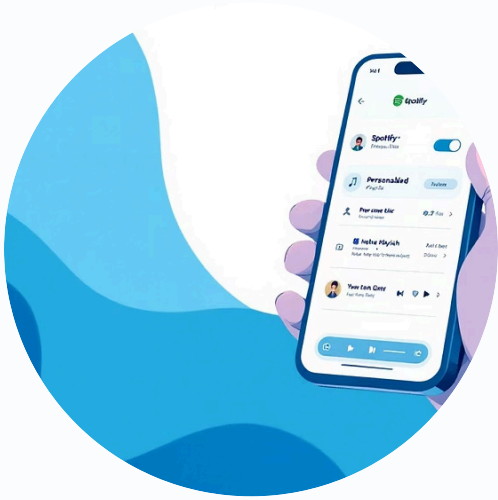
Social media platforms use AI to curate your feed, showing you content they think you'll like based on your interactions. Snapchat filters use computer vision AI to detect your face and overlay effects.



ChatGPT / Gemini

These conversational AI models can answer questions, write stories, explain concepts, and even help with homework. They are trained on vast amounts of text and learn to generate human-like responses.

AI in Everyday Life



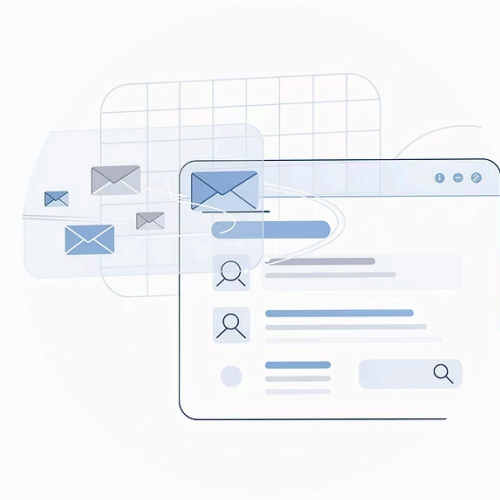
Spotify / Apple Music

Music streaming services create personalized playlists (like "Discover Weekly") by analyzing your listening habits and comparing them to others with similar tastes.



Face Unlock on Smartphones

Many phones use AI-powered facial recognition to securely unlock when they see your face. The AI learns to recognize you even if you wear glasses or grow a beard.



Spam Filters in Email

Services like Gmail use AI to detect unwanted emails (spam) by learning patterns from millions of messages. They get better over time as you mark emails as spam or not spam.



History of AI

British mathematician Alan Turing published a paper asking, "Can machines think?" He proposed a test (now called the Turing Test) to determine if a machine could exhibit intelligent behavior indistinguishable from a human.

History of AI

In 1956, a group of scientists met at Dartmouth College in the United States. This is where the field of **Artificial Intelligence officially got its name**. They were very excited and believed machines could be made to think like humans. It turned out to be much harder than they expected - **but they started the journey!**



History of AI

1950s – The Beginning

Alan Turing asked "Can machines think?" The Dartmouth workshop coined the term "**Artificial Intelligence**" – and a new field was born!

1980s–90s – Expert Systems & Chess

AI got smarter! In 1997, IBM's **Deep Blue** beat chess champion Kasparov – shocking the world.

2010s – Deep Learning Revolution

AlphaGo beat the world champion at Go – a game many thought was too complex for machines.

1960s–70s – Early Experiments

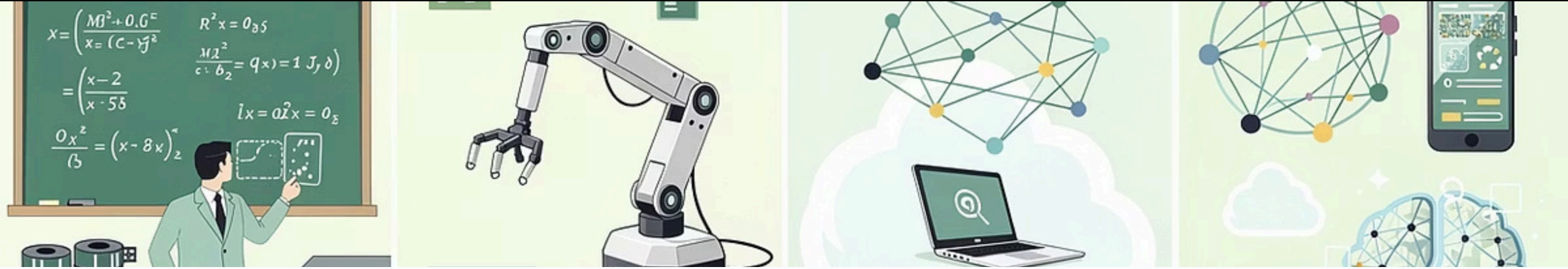
Programs like ELIZA could chat with humans. But progress slowed down, leading to the first "**AI winter**" when funding dried up.

2000s – Machine Learning Rises

Speech recognition and recommendation systems became part of daily life. AI was learning faster than ever.

2020s – Generative AI Explodes 🚀

ChatGPT, DALL-E, and more arrived. AI can now write stories, create art, and have conversations. **YOU ARE HERE!**



Key Takeaway from AI History

AI has been around for a long time, but recent advances have made it much more powerful and accessible.

How AI and Humans Differ - Learning & Creativity

AI is impressive, but it's *very different* from the human brain. Let's compare!



Human Learning

Learns from just **a few examples** and generalizes easily. Show a kid 3 pictures of dogs and they understand "dog" right away!



Human Creativity

Creates **truly original** ideas fueled by emotion, imagination, and lived experience. You dream up things nobody has seen before!



AI Learning

Needs **thousands or millions** of examples to learn. Struggles to transfer knowledge from one task to another.



AI "Creativity"

Remixes patterns from its training data. It doesn't have original thoughts — it recombines what it has already seen.

💡 Humans learn quickly and create genuinely new things. AI learns slowly and remixes what it has seen.

How AI and Humans Differ - Common Sense & Emotions



Common Sense

- **Humans:** Built through living in the real world. You know a glass will break if dropped, even if nobody told you.
- **AI:** Lacks real-world experience. It can make absurd mistakes because it doesn't truly understand how the world works.



Emotions

- **Humans:** We feel joy, sadness, empathy, and love. Our emotions shape how we connect with others.
- **AI:** Can simulate empathy in words, but **feels nothing**. It has zero emotions or consciousness.



Consciousness

- **Humans:** We are self-aware with subjective experiences. We know that we exist and can reflect on our thoughts.
- **AI:** No awareness at all, just mathematical patterns processing data. It doesn't know it exists.

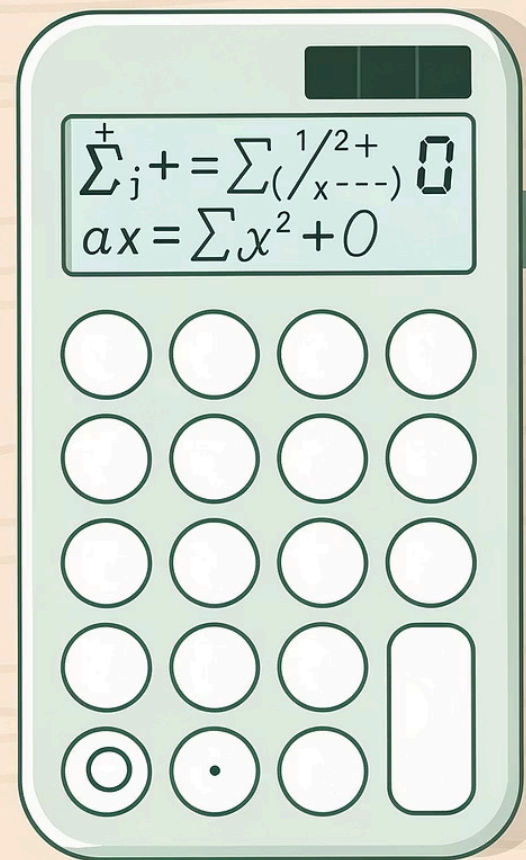
AI vs Human Intelligence

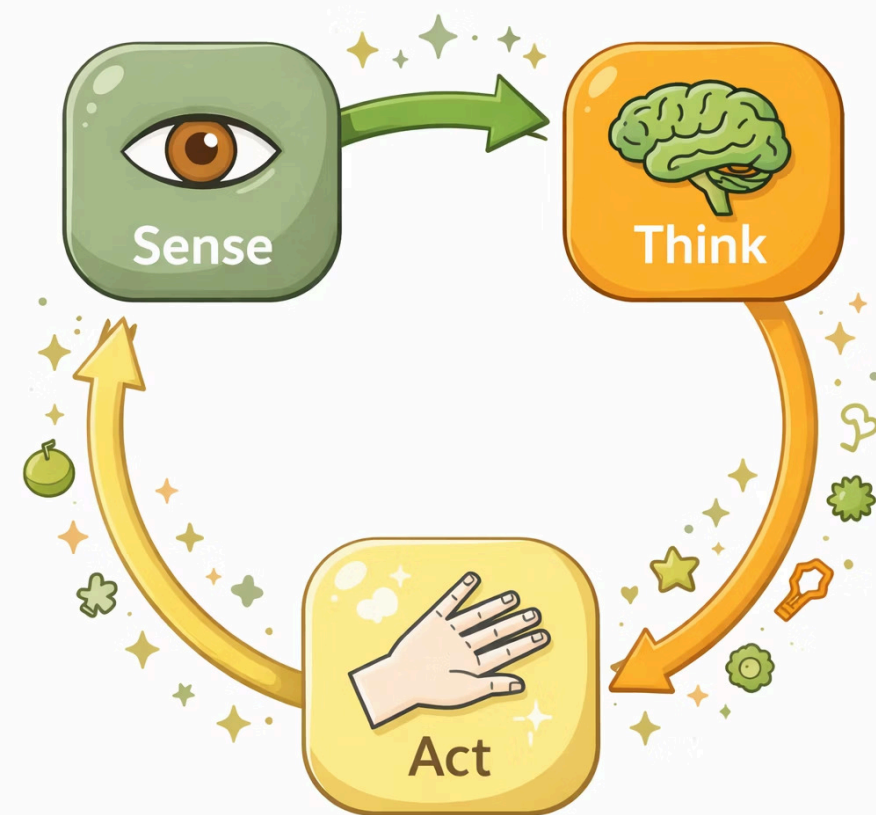
Aspect	Human Intelligence	Artificial Intelligence
Learning	Humans can learn from just a few examples and apply that knowledge to many different situations. We generalize easily.	AI typically needs thousands or millions of examples to learn a task, and it struggles to apply what it learned to new situations (transfer learning is limited).
Creativity	Humans create truly novel art, music, and ideas, often driven by emotion and experience.	AI generates content by remixing patterns it has seen. It doesn't have original ideas or emotions, though it can mimic creativity.
Common Sense	We build common sense through living in the world. We know that if you drop a glass, it might break.	AI lacks real-world experience. It may make absurd mistakes that any human would avoid, because it doesn't truly "understand" the world.
Emotions	Humans feel joy, sadness, empathy. These emotions influence our decisions and interactions.	AI can simulate empathy (e.g., a customer service bot saying "I understand you're frustrated"), but it feels nothing. It's just following patterns.
Speed	Humans are relatively slow at processing data, but we can multitask and adapt quickly.	AI can process vast amounts of data in milliseconds, but it's typically focused on one task at a time.
Flexibility	We can switch between completely different tasks effortlessly – from cooking to solving a math problem.	Most AI is narrow: it can only do what it was trained for. A chess AI cannot drive a car.
Consciousness	Humans are self-aware; we have subjective experiences.	No AI today has consciousness or self-awareness. It's a tool, not a sentient being.

The Calculator Analogy

Analogy: Think of AI as a calculator for thinking. A calculator does math faster than any human, but it doesn't understand what numbers mean. Similarly, AI can process information at incredible speed, but it lacks true understanding.

This distinction is crucial for responsible use. We should not expect AI to have human judgment, and we must always double-check its outputs. In later sessions, we'll see examples of AI making confident mistakes (hallucinations) and why that happens.





The Sense–Think–Act Framework

AI systems, from voice assistants to self-driving cars, follow the same three-stage process. Understanding this framework helps you see how *any* AI system works!

This simple model is your secret decoder ring for understanding AI. Whenever you encounter a new AI tool, just ask yourself: **What does it sense? How does it think? What does it do?** 🔍

SENSE - How AI Perceives the World

AI first needs to gather information from its environment. This is called perception. Depending on the AI, sensing can involve:

- Cameras to see images or video (like in face recognition or self-driving cars).
- Microphones to hear sound (voice assistants).
- Sensors to detect temperature, motion, location (in robots or smart devices).
- Text input - the words you type into ChatGPT or Google Search.

The data collected during sensing is raw and unstructured. It might be pixels, sound waves, or strings of text.



SENSE - How AI Perceives the World

In the **SENSE** stage, AI collects data - the raw information it needs to start working. Think of it like AI's eyes, ears, and fingertips! Here's how it gathers information:



Cameras

To see images and video — used in face recognition, self-driving cars, and photo apps



Microphones

To hear sound — used by voice assistants like Siri, Alexa, and Google Assistant



Sensors

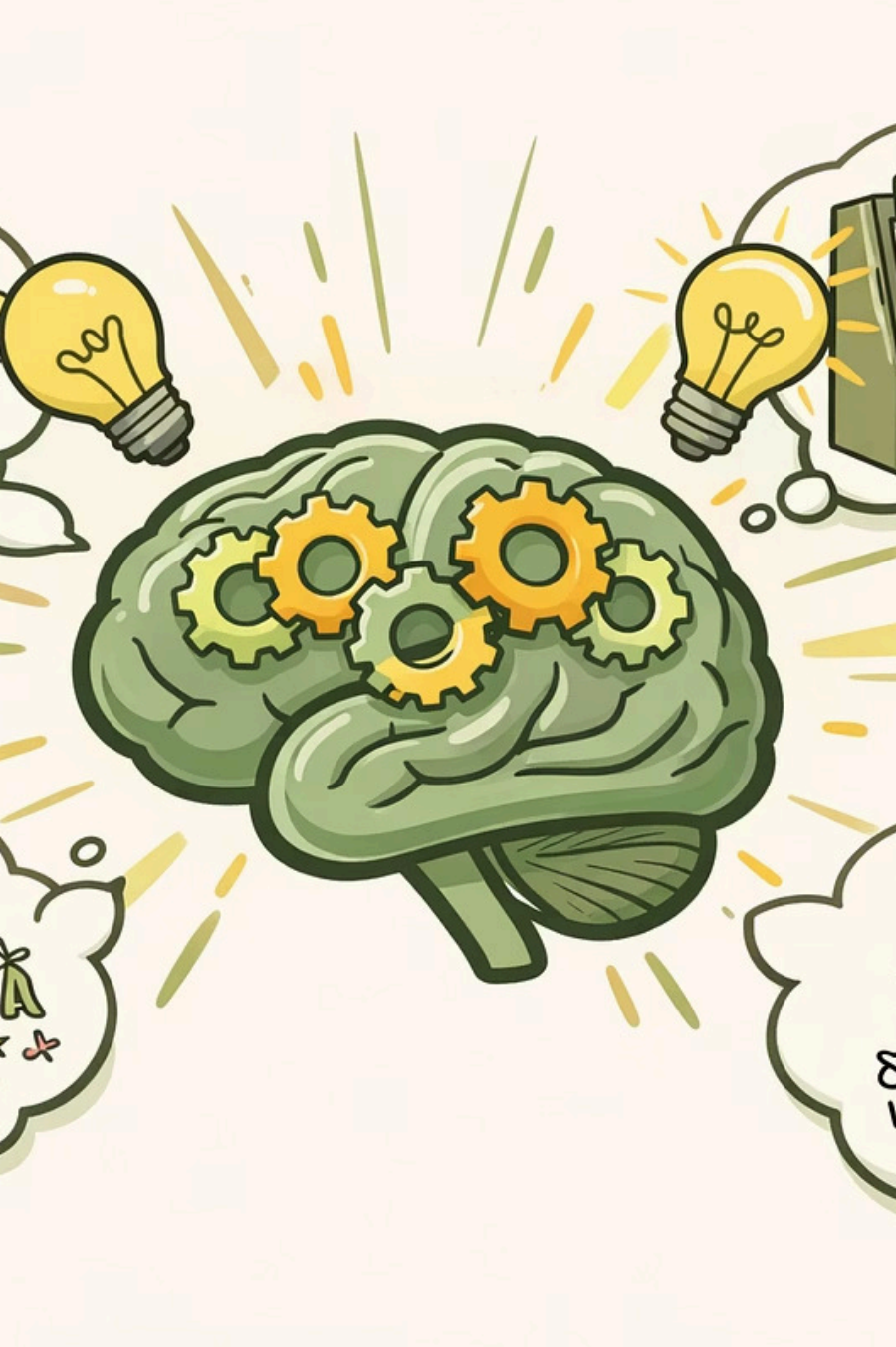
To detect temperature, motion, and location — used in robots, smart homes, and fitness trackers



Text Input

Words you type — used by search engines, chatbots, and writing assistants

The data collected is **raw** — pixels, sound waves, letters — and becomes the input for the next stage: thinking! 🧠



THINK - How AI Process Data

Once the AI has data, it needs to make sense of it. This is the "thinking" stage, where the AI uses algorithms and models to:

- Recognize patterns (is that a cat or a dog?)
- Make predictions (what movie will you like?)
- Decide on a course of action (should the car brake?)
- Generate new content (write a poem).

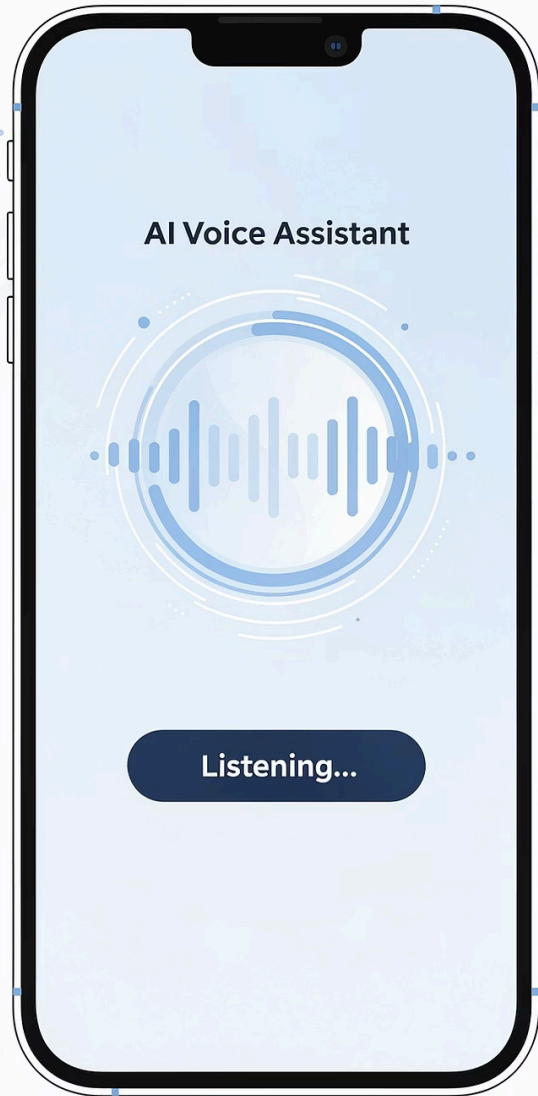
The "thinking" is done by a model - a mathematical representation trained on vast amounts of data. For example, a language model like ChatGPT has been trained on billions of sentences and can predict the next word in a sequence.

ACT - How AI Output Results

Finally, the AI produces an output based on its thinking. The action could be:

- Displaying text - an answer to your question.
- Speaking - a voice assistant responding.
- Moving - a robot arm picking up an object.
- Changing something on screen - a recommendation appearing.





Think and Act - Processing and Output

THINK

AI processes data using algorithms and models to:

- **Recognize patterns** - Is that a cat or a dog?
- **Make predictions** - What will you watch next?
- **Make decisions** - Should the car brake now?
- **Generate content** - Write a story or draw a picture

ACT

AI produces an output you can see, hear, or interact with:

- Displaying text on a screen
- Speaking an answer out loud
- Moving a robot or steering a car
- Changing what appears on your screen
- Sending you a notification or alert



Real-World Examples of Sense-Think-Act



Voice assistant (Siri)

- Sense (microphone captures sound)
- Think (converts sound to text, interprets intent, decides answer)
- Act (speaks response)



Self-driving car

- Sense (cameras, radar, lidar detect road, cars, pedestrians)
- Think (AI decides when to brake, steer, accelerate)
- Act (car moves accordingly).



Netflix recommendation

- Sense (collects data on what you've watched)
- Think (compares with patterns from other users)
- Act (displays recommended videos)

Activity Time





Let's Play: "AI or Not AI?"

Time for a game! We'll look at different technologies, and for each one, you'll tell us: **Is it AI or Not AI?** And most importantly - *why* do you think so?



Does It Learn?

Does it learn from data and get better over time?



Does It Improve?

Does it change its behavior based on new information?



Does It Predict?

Does it make predictions or decisions on its own?

Calculator

Netflix
recommendation

Traffic light

Google Maps traffic
prediction

Spell check

Microwave with
preset buttons

Snapchat filters

Amazon purchase
suggestions

Calculator

Not AI (follows fixed rules; does not learn or adapt).

Netflix recommendation

AI (learns from your viewing history; improves over time).

Traffic light

Not AI (operates on a timer or simple sensors; no learning).

Google Maps traffic prediction

AI (uses real-time data and historical patterns to predict traffic).

Spell check

Can be AI (modern spell checkers learn from corrections) or rule-based (older ones). Discuss: if it learns from you, it's AI.

Microwave with preset buttons

Not AI (just follows programmed settings).

Snapchat filters

AI (detects face, tracks movements; uses computer vision).

Amazon purchase suggestions

AI (analyzes your browsing and purchase history).



AI Myths vs. Reality

There are a lot of wild ideas about AI floating around. Let's separate **fact from fiction!** 🕵️

❌ **MYTH:** "AI is only robots"

✅ **REALITY:** AI is *software*, code running on computers. Robots are hardware that *may* use AI. Most AI lives quietly in the cloud or on your phone, not in a metal body!

❌ **MYTH:** "AI will take over the world"

✅ **REALITY:** Today's AI is *narrow*, it does specific tasks really well but has no consciousness, goals, or desire to "take over." It's a tool, not a villain!

❌ **MYTH:** "AI is always correct"

✅ **REALITY:** AI makes mistakes, sometimes very confidently! It can give wrong answers that *sound* right. Always double-check important information. 🔍

AI Myths vs. Reality

✗ **MYTH:** "AI is brand new"

✓ **REALITY:** AI research began way back in the **1950s**! We've been using AI for decades in spam filters, recommendations, and search engines. It's not new, it's just gotten much better recently.

✗ **MYTH:** "AI has feelings"

✓ **REALITY:** AI can *simulate* empathy and kindness in its words, but it **feels nothing**. No joy, no sadness, no consciousness. It's all mathematical patterns behind the scenes.

✗ **MYTH:** "AI understands like humans"

✓ **REALITY:** AI processes patterns but **lacks true understanding**, common sense, and real-world life experience. It's great at matching patterns, not at truly "getting it."

AI is powerful, but it's **not magic** and **not human**. Understanding its limits helps us use it wisely! 🌟

Two Types of AI : Narrow AI vs General AI

Narrow AI (Today) 🎯

Designed for **one specific task**. Examples include chess AI, face recognition, and ChatGPT. *All AI that exists today is narrow AI.*

General AI (Future) 🧙

Could perform **any** intellectual task a human can. But it **does not exist yet**. It's still a long-term research goal and a hypothetical idea.

Not all AI is created equal. Scientists talk about two main types and one of them **doesn't even exist yet!**



Think of It This Way...

Narrow AI is like a master chef who can cook anything, but *only* in the kitchen. Put that chef in a garage to fix a car, and they're completely lost!

General AI would be like a person who can cook, drive, paint, do math, play sports, versatile and adaptable at everything. But remember: **General AI doesn't exist yet!**

AI is a Tool, Not a Replacement

Think of AI like a **calculator**: a calculator does math faster than you, but *you* decide what problem to solve, and *you* decide if the answer makes sense. AI works the same way!

AI assists, you create

AI can help brainstorm and draft ideas, but your creativity drives the project forward.

AI suggests, you decide

AI offers options and recommendations, but you choose what's right for the situation.

AI drafts, you polish

AI can create a rough version, but your judgment and personal touch make it great.

AI can help, but you are responsible

Whatever AI produces, *you* are the one who checks, edits, and takes responsibility.





Hands-On Activity

Now it's time to roll up your sleeves and experience AI for yourself!

In the next few slides, you'll become an AI storyteller using AI as your creative partner to build an amazing story together!

Get ready ✨



Purpose: See how AI helps with creative tasks, learn that *your input shapes the output*, and have fun creating something totally unique!

Meet Google Storybook

Google Storybook is an AI tool designed specifically to help you create stories. Here's how it works:

- 1 You Start**
Type about your idea to kick off your story
- 2 AI Reads**
The AI reads your sentence and adds the next part
- 3 AI Builds**
The AI continues building on your ideas and creates your storybook

It's like having a **creative partner** who never runs out of ideas, builds on whatever you suggest, and sometimes surprises you with unexpected twists! 🌟



Hands-On Activity: Create a Story with AI 🖋️

Today you'll experience AI as a **creative partner**!

1 **Open Google Storybook**

We'll use this AI storytelling tool together as a class

2 **Start a Story**

You'll begin with your own creative opening sentence

3 **Watch AI Add More**

See how the AI builds on your ideas and adds the next part

4 **Guide the Direction**

YOU decide where the story goes, the AI follows your lead!

5 **Finish Your Masterpiece**

End with a complete story created by YOU + AI together 🎉

Your Turn to Create! 🌈

Your Task

Continue our story together!
You'll suggest what happens next at each step and watch how the AI responds to your ideas.

Be Creative

There are NO wrong ideas, AI can work with anything you throw at it! If the story goes somewhere unexpected, that's part of the fun.

You're in Control

Remember: YOUR ideas guide the direction. The AI is your creative partner, but YOU are the lead storyteller.

Let's create something amazing together! ✨



What We Learned Today

Wow, we covered a LOT today! Let's recap the big ideas from our AI adventure: 🎓

✅ What is AI

Computers performing tasks that normally need human thinking — like **learning, recognizing patterns,** and **understanding language**. It's computer science at its cleverest!

✅ AI in Daily Life

You already use AI every day! Netflix, YouTube, voice assistants, Google Maps, Spotify, and so many more apps rely on AI behind the scenes.

✅ History of AI

AI research began all the way back in the **1950s** and has evolved over **70+ years** — from early chatbots to today's generative AI!

✅ AI vs. Human Intelligence

AI learns differently from us, has **no feelings**, and lacks common sense. It's brilliant at patterns, but not at true understanding.

What We Learned Today

✓ How AI Works

Sense (gather data) → **Think** (process it) → **Act** (produce output). This framework explains every AI system!

✓ Common Myths - Busted!

AI is not just robots, not always correct, not brand new, has no feelings, and doesn't truly understand the world like we do.

✓ Narrow vs. General AI

All AI today is **narrow** (one task at a time). General AI — the do-everything kind — is still a future dream that doesn't exist yet.

✓ AI as a Tool

AI assists, **YOU** create. AI suggests, **YOU** decide. **YOU** are always in charge and responsible for the final result.

☀ **The Most Important Message:** AI is a powerful tool that enhances human capabilities — but it does **NOT** replace human judgment, creativity, or responsibility.

Your Homework: Create Your Own AI Story

Time to practice at home! Here's your assignment:

1 Get Started

Go to **Google Storybook** (link shared in chat or parent email)

2 Start Your Story

Choose your own opening sentence or let the AI begin for you!

3 Build It Together

Let AI add sentences and create final storybook for you

4 Save Your Work

Screenshot it, copy the text, or grab the shareable link

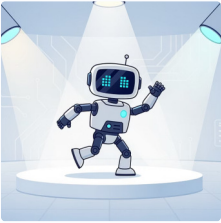
5 Bring It Back!

Bring your story to our next session so we can share our creations! 🎉

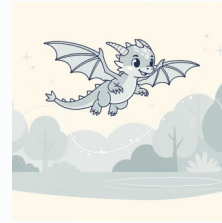
 **Time estimate:** About 15 minutes. Have fun with it - try different directions and see what the AI comes up with!

Need an Idea? Try One of These! 💡

If you're staring at a blank screen, no worries! Pick one of these story starters to get your creative juices flowing:



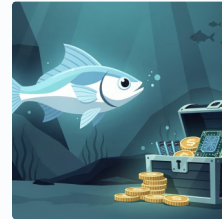
"Once upon a time, there was a robot who loved to dance."



"In a magical forest, a young dragon learned to fly."



"Maya and her friends discovered a secret message in a bottle."



"Deep in the ocean, a curious fish found a glowing treasure."



"Max built a machine that could talk to animals."

Upcoming Session : AI Concepts & Terminology

Today we explored *what AI can do*. Next session, we'll learn the **special words** that experts use to talk about AI - the building blocks of the **AI**!



Algorithm

The step-by-step recipe AI follows



Data

The information AI learns from



Model

The trained recipe ready to use



Training

Learning from lots of examples



Prediction

Making a smart guess based on patterns

These are the building blocks we'll use throughout the rest of our AI journey. **See you next time!**  